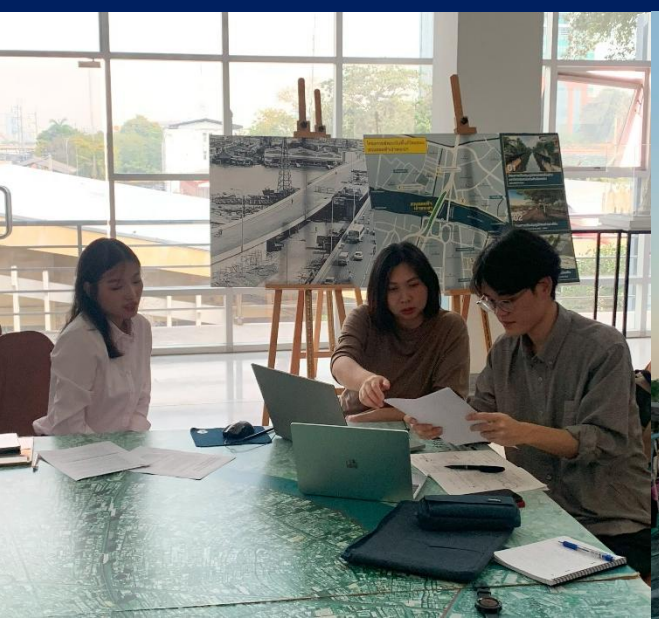
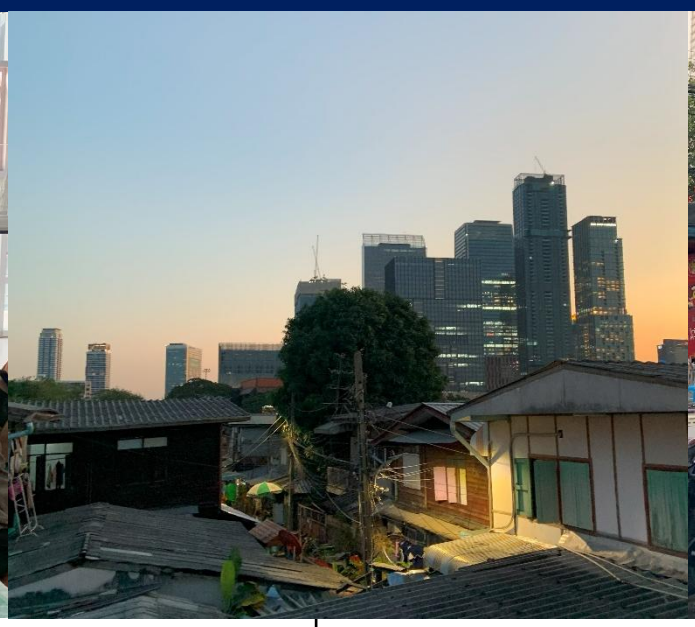
Impact of Urban Informal Transport on Gentrification near Rail Stations: Case Study in Bangkok

インフォーマル都市交通がジェントリフィケーションの進行に与える影響：タイ・バンコクにおける事例分析

Shota EMOTO
柄本 將太


1 Background

Transport Development and Gentrification

- Intensive public transport development causes the issue of gentrification, especially in developing cities like Bangkok

Informal Transport and Gentrification

- Informal transport serves as a means for low-income people to access rail stations

Research question

Does urban informal transport mitigate gentrification near rail stations?

2 Goals

This study aims to:
(1) Estimate empirically the mitigation effect of informal transport to gentrification in Bangkok, Thailand;
(2) Discuss policy implications

3 Methods

Causal Analysis

- Data collection using Google Maps Street View
- Estimation of Average Treatment Effect on the Treated (ATT) with propensity score matching

Field Surveys

- Interviews with local experts and professionals
- Field visits in Bangkok
- Identification of the impact mechanism

Discussion on policy implications

Assumptions in Causal Analysis

- Treatment group: Properties in rail station areas located within 50m from motorcycle taxi stations
- Control group: Properties in rail station areas located further than 50m from motorcycle taxi stations

5 Hypotheses

Hypothetical Mechanism

Access to informal motorcycle taxi (MT) in rail station areas

Low-income people are benefited by rail development

Increase in income and quality of life for low-income people

Increase in income level for local residents

Stable life condition of lower-income citizens

Rise in price of the products and services in the area

Mitigation of residential gentrification

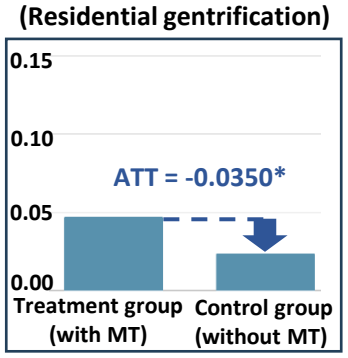
Promotion of commercial gentrification

6 Results

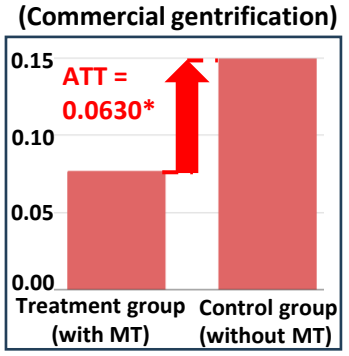
Impacts of MT stations on gentrification near rail stations

- ATTs were estimated for two types of gentrifications

ATT estimation result (Residential gentrification)



ATT estimation result (Commercial gentrification)



Treatment group N=254

Control group N=254

* 5% significance

Findings

1. Residential gentrification was mitigated by MT access
2. Commercial gentrification was promoted by MT access

7 Conclusions

- MTs may help mitigate residential gentrification in rail station areas
- Informal transport should be smartly integrated with transport planning for mitigating gentrification.

Shota EMOTO
Final Study of undergraduate program in 2025
Supervisor: Prof. Hironori Kato

